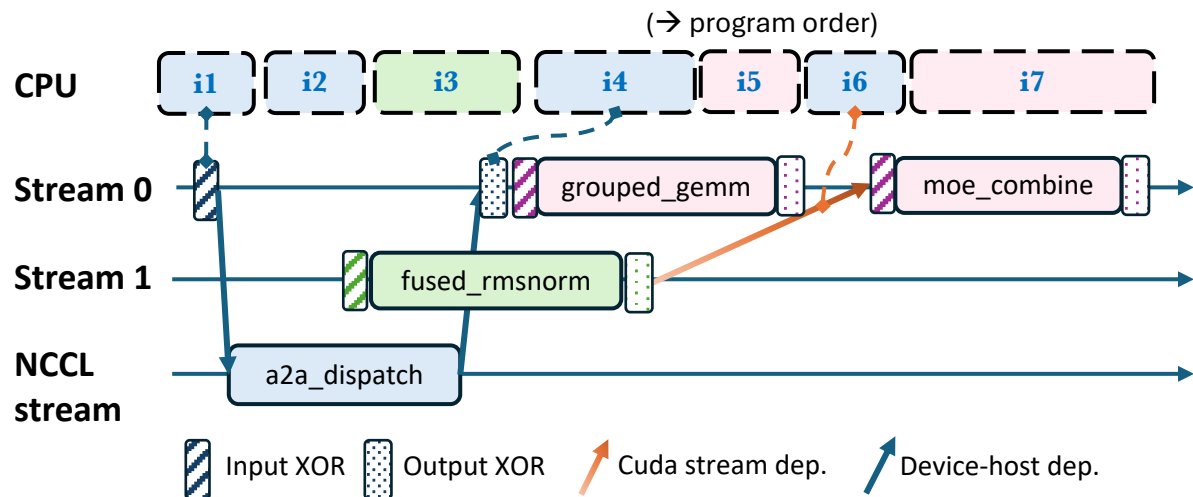


(a) Preflight + guarded execution



```

i1 x0, h = all_to_all_dispatch(x0, ...)
i2 with torch.stream(s1):
i3   y1 = fused_rmsnorm(residual)
i4 h.wait()
i5 x1, h = grouped_gemm_exprts(x0, ...)
i6 torch.wait_stream()
i7 out = moe_combine(x1, y1)
    
```

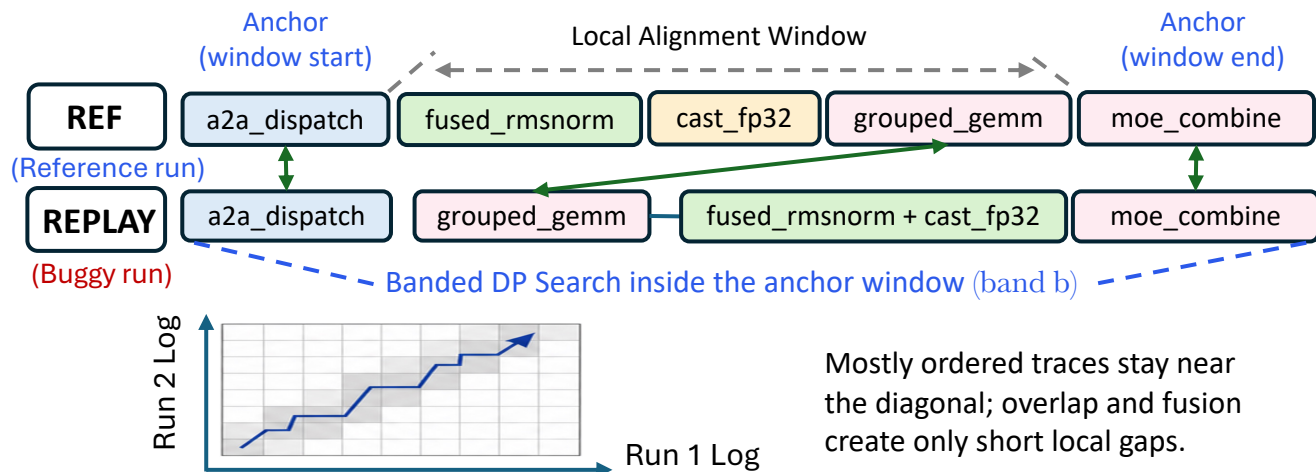
Uninstrumented
Program

```

i1  x0, h = wrapped_all_to_all_dispatch(x0, ...) 
i2 with torch.stream(s1):
i3    y1 = wrapped_fused_rmsnorm(residual) 
i4 h.wait()
i5  x1, h = wrapped_grouped_gemm_exprts(x0, ...) 
i6 torch.wait_stream()
i7  out = wrapped_moe_combine(x1, y1) 
    
```

XOR
probes
installed

(b) Schedule-tolerant alignment



(c) Certified-prefix localization

Aligned boundary pairs (in order)

